

AKHISAR, Turkey

WMO: 171840

Lat: 38.9114N

Lon: 27.8230E

Elev: 305

StdP: 14.53

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.0	29.5	20.6	15.6	31.4	23.3	17.8	33.1	18.6	44.0	15.7	43.4	2.0	350	0.405

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.8	101.0	70.8	98.1	70.7	95.4	70.4	75.2	94.0	73.6	91.8	72.2	90.1	5.6	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.3	108.9	85.2	67.6	102.6	83.1	66.1	97.3	81.7	38.8	94.0	37.4	91.6	36.1	90.1	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.8	13.3	11.5	DB	21.9	106.1	3.5	2.8	19.4	108.2	17.3	109.8	15.3	111.4	12.8	113.4	(4)
(5)			WB	20.8	77.4	3.6	2.4	18.2	79.2	16.1	80.6	14.1	82.0	11.5	83.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.7	43.7	46.8	51.5	59.3	68.9	77.4	82.6	82.5	74.6	64.8	53.7	45.9		
	DBStd	15.03	6.56	6.76	6.34	6.46	5.49	5.25	3.72	3.73	5.25	5.83	6.63	6.82		
	HDD50	610	213	128	59	7	0	0	0	0	0	1	37	165		
	HDD65	2817	660	510	419	192	26	1	0	0	2	75	341	591		
	CDD50	5258	19	38	106	286	587	822	1009	1008	737	459	148	39		
	CDD65	1986	0	0	1	20	148	372	544	543	288	68	2	0		
	CDH74	23214	0	2	19	340	1699	4345	6823	6510	2884	571	21	0		
	CDH80	11869	0	0	2	92	646	2230	3852	3591	1317	138	1	0		
(14) Wind	WSAvg	3.9	3.3	3.7	4.2	3.6	3.7	4.5	5.9	5.7	3.7	3.3	2.4	2.8		
(15) Precipitation	PrecAvg	22.3	3.1	3.2	2.6	2.1	1.5	0.6	0.2	0.2	0.8	1.7	2.7	3.5		
	PrecMax	31.2	7.8	8.0	5.7	5.4	8.1	1.9	0.9	0.9	2.7	6.4	7.2	8.6		
	PrecMin	12.7	0.1	0.4	0.7	0.4	0.1	0.0	0.0	0.0	0.0	0.1	0.0	0.0		
	PrecStd	5.0	2.0	2.0	1.5	1.2	1.6	0.6	0.2	0.2	0.8	1.5	1.7	2.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.2	69.7	76.3	88.4	94.4	103.2	106.1	103.2	100.1	88.6	77.2	67.6		
		MCWB	54.9	56.4	58.9	62.3	66.3	71.3	72.5	71.2	68.8	66.7	61.8	57.3		
	2%	DB	61.3	65.2	71.4	82.4	90.7	98.1	101.2	100.2	94.8	83.6	72.1	63.3		
		MCWB	53.3	54.7	56.2	62.1	66.1	70.1	71.2	71.0	68.6	64.6	59.9	56.1		
	5%	DB	57.7	62.0	68.0	77.6	87.1	94.4	98.1	97.4	91.0	80.1	68.7	59.7		
		MCWB	51.3	52.9	55.1	60.5	65.3	69.4	70.7	71.1	67.4	63.5	58.1	54.2		
	10%	DB	54.2	58.4	64.2	73.4	83.4	91.3	95.2	94.7	87.7	76.6	65.2	56.7		
		MCWB	49.4	51.2	53.7	58.8	64.2	68.3	70.7	71.2	66.7	62.4	57.0	51.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.3	57.6	60.9	65.8	70.2	74.5	77.3	77.6	72.7	69.4	63.9	58.7		
		MCDB	63.0	66.5	73.2	81.8	87.9	96.4	97.0	96.0	92.2	82.5	73.2	64.6		
	2%	WB	53.8	55.3	57.7	63.4	68.3	72.4	75.3	75.5	70.8	66.9	61.2	56.5		
		MCDB	60.4	63.9	68.0	79.0	85.8	92.7	94.9	93.1	88.9	78.2	69.1	62.0		
	5%	WB	51.7	53.5	56.0	61.3	66.6	70.9	73.6	73.9	69.5	65.4	59.4	54.7		
		MCDB	56.6	60.6	65.5	75.1	83.1	90.0	92.0	91.1	86.2	76.2	66.5	59.4		
	10%	WB	49.8	51.7	54.3	59.3	65.1	69.7	72.1	72.5	68.2	63.9	57.6	52.6		
		MCDB	53.8	57.9	63.0	71.5	80.4	87.8	90.2	89.4	83.6	74.2	64.0	56.3		
(35) Mean Daily Temperature Range	MDBR		14.9	16.9	19.9	22.7	24.7	25.2	24.8	24.6	24.6	21.4	19.1	14.9		
	5% DB	MCDBR	18.8	23.4	27.5	30.8	31.5	30.1	30.2	28.7	30.0	26.7	24.4	16.8		
		MCWBR	12.3	14.0	14.5	14.1	12.4	10.4	9.2	8.2	10.3	11.2	13.4	10.9		
	5% WB	MCDBR	17.4	20.9	23.8	27.9	27.1	26.6	25.5	24.2	25.6	22.2	20.8	15.8		
		MCWBR	11.8	13.0	13.3	13.7	11.6	9.8	9.1	8.3	9.7	10.5	12.2	10.7		
(40) Clear-Sky Solar Irradiance	taub		0.356	0.392	0.433	0.474	0.449	0.413	0.407	0.414	0.403	0.398	0.371	0.358		
	taud		2.354	2.224	2.108	2.026	2.164	2.304	2.320	2.292	2.311	2.309	2.342	2.363		
	Ebn at Noon		254	259	259	256	266	275	275	271	266	254	246	244		
	Edh at Noon		29	37	46	53	47	41	40	40	37	34	29	27		
(44) All-Sky Solar Radiation	RadAvg		622	844	1269	1668	2056	2339	2463	2176	1694	1173	791	571		
	RadStd		81	101	107	169	128	158	64	78	108	79	82	84		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) +1.30	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (15 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	-166	N/A	N/A

Nomenclature: See separate page